Hongzhou Luan

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RESEARCH INTERESTS

I work on maximizing AI's impact on healthcare for the underprivileged majority. My research addresses two connected problems in clinical AI deployment: extracting signal from informative, not-at-random missingness in healthcare and wearable data, and closing the gap between closed-world evaluation (fixed benchmarks, complete ground truth) and the open-world clinical environments where AI systems actually operate. This work is grounded in a broader commitment to equitable and safe AI deployment, including bias analysis in clinical algorithms and AI policy.

EDUCATION

University of California, Berkeley – UC San Francisco 2023 – Present
PhD, Computational Precision Health
Advisors: Prof. Ida Sim (primary), Prof. Ahmed Alaa
Funding: JupyterHealth

University of Oxford 2018 – 2022
MEng, Engineering Science
Jardine Scholar (full four-year scholarship)

PUBLICATIONS

[1] **Hongzhou Luan**, Leeor Hershkovich, Ida Sim, Jessilyn Dunn. "Wearable Missingness as a Behavioral Signal: Detecting Wear Pattern Anomalies Around Cardiovascular Visits." *ICML SD4H Workshop*, ***Spotlight***, 2026.

[2] Anna Zink, **Hongzhou Luan**, Irene Y. Chen. "Access to care affects electronic health record reliability and AI-driven disease prediction." ***Nature Health***, 2026. doi:10.1038/s44360-026-00054-9

[3] Hugo Campos Jr, Daniel Wolfe, **Hongzhou Luan**, Ida Sim. "Generative AI as Third Agent: LLMs and the Transformation of the Clinician-Patient Relationship." *Journal of Participatory Medicine*, 2025.

[4] Anna Zink, **Hongzhou Luan**, Irene Y. Chen. "Access to care improves EHR reliability and clinical risk prediction model performance." *ML4H*, 2024.

RESEARCH EXPERIENCE

Wearable Data Missingness and Acute Event Prediction 2025 – Present
Advisors: Prof. Jessilyn Dunn (Duke University), Prof. Ida Sim

- Analyzed Fitbit data linked to EHR in the All of Us dataset to investigate whether shifts in wearable data missingness patterns precede acute cardiovascular events.
- Developed an anomaly detection algorithm that identifies significant changes in missingness patterns around acute healthcare visits, treating non-wear as a behavioral signal rather than data to impute.

AI Scribes Evaluation in Clinical Practice 2025 – Present
Advisors: Prof. Ahmed Alaa, Prof. Ida Sim, Prof. Paul Tang (Stanford)

- Leading the **first systematic evaluation for clinical harms** of AI medical scribes, collecting and analyzing 400 clinical encounters including raw audio, AI-generated notes, and physician-edited final notes.
- Investigating how clinician editing behaviors vary by user experience and clinical context, using edit patterns as a naturalistic signal for identifying systematic AI failure modes.

Dermatology Triage for Skin Cancer Screening 2024 – Present
Advisors: Prof. Katrina Abuabara, Prof. Meghan Shan **Accepted at SAIL 2026**

- Developing a novel triage algorithm using patient-uploaded facial images linked to demographic and clinical data. Independently filed the successful IRB for this project.
- Addressing extreme class imbalance and demographic representation gaps through multiple demographic-aware balancing strategies, achieving a 47% reduction in sensitivity disparity across racial and ethnic groups.
- Departing from traditional melanoma AI that requires high-quality dermoscopic images, instead leveraging noisy, real-world images already captured in clinical workflows.

Missingness-Avoidant Neural Networks for Tabular Data 2025 – Present

Advisor: Prof. Fredrik Johansson (Chalmers University of Technology)

- Designing attention-based architectures that predict from observed features only, removing the dependence on imputation, which assumes missing-at-random data and lacks clinical interpretability.
- Targeting the informative, not-at-random missingness common in healthcare data, where imputation obscures behavioral signal and undermines clinician trust.

AI Policy Research and Legislative Drafting 2025 – 2026

UC Berkeley CITRIS Policy Lab, California Initiative for Technology and Democracy (CITED)

- Conceived and led the Grades for Responsible AI in Daily Environments (GRADES) framework for AI safety policy.
- Led a team of three to draft a legislative bill in partnership with CITED, and presented the bill to key state legislative staff members in Sacramento.

AWARDS & HONORS

Spotlight, ICML SD4H Workshop 2026

Jardine Scholarship, Jardine Foundation 2018 – 2022

Full scholarship covering four years at Oxford with an annual stipend of £10,300, awarded to exceptional students from Asia demonstrating academic excellence and leadership potential.

MENTORSHIP & SERVICE

Volunteer Graduate & Undergraduate Admissions Counselor 2019 – Present

Provide ongoing mentorship for first-generation students and women in STEM navigating undergraduate and graduate school applications. Mentees have been admitted to Stanford, University of Pennsylvania, Oxford, Cambridge, and other top institutions, with many receiving full scholarships.